

Question Paper

Exam Date & Time: 26-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: HUM 3021/HUM_3021 ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data may be suitably assumed.

Interest factor table is provided for others use formula.

Formula book is attached with the paper.

- 1A) Mr. Ramesh initiated a structured investment plan on March 1, 2020, by depositing a fixed amount (4) each month into a savings account that earns 10% nominal interest compounded monthly. His objective was to build long-term savings while maintaining liquidity through regular quarterly withdrawals. Each monthly deposit amounts to ₹2,00,000. These deposits were made on the 1st of every month from March 1, 2020 to October 1, 2025. Simultaneously, the account provided quarterly withdrawals to meet short-term financial needs. The first withdrawal of ₹5,00,000 occurred on July 1, 2020, and similar withdrawals continued every three months until October 1, 2025. Calculate the amount available in the account as of October 1, 2025, after the deposit and withdrawal on that day.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Alex is considering buying property that comprises a plot of land and a shop. The land is predicted (3) to increase in value by ₹30,00,000 over five years and shop has no salvage value. The shop can be rented out immediately for a net annual revenue of ₹3,50,000. Annual maintenance expenditures are estimated to start at ₹50,000 in the second year and grow by 2,000 annually. If Alex wants a 20% return on his investment, how much can he pay for this property now?

20%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.200	.8333	1.0000	1.2000	1.000	0.833	0
2	1.440	.6944	.4545	.6545	2.200	1.528	0.455
3	1.728	.5787	.2747	.4747	3.640	2.106	0.879
4	2.074	.4823	.1863	.3863	5.368	2.589	1.274
5	2.488	.4019	.1344	.3344	7.442	2.991	1.641
6	2.986	.3349	.1007	.3007	9.930	3.326	1.979

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Ravi is planning for his retirement. He has 10 more years until retirement. He decides to deposit (3) ₹12,000 at the end of the first year and will decrease the amount by ₹700 each year for the next 7 years. If the savings account offers an interest rate of 10% per annum, what will be the total accumulated amount at the time of retirement?

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) On 31st December 2011, two machines originally purchased on 1st January 2008 had to be discarded due to irreparable damage. At the time of purchase, the first machine cost was ₹15,000 and had an estimated useful life of 10 years, while the second machine cost was ₹12,000 with an estimated life of 8 years. Both machines were sold for ₹5,000 each at the time of disposal. The task is to determine the resulting capital gain or sunk cost under two depreciation approaches: the Declining Balance Method (rate of depreciation is 10%) and the Double Declining Balance Method for the two machines. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) An Indian consulting firm is considering acquiring an all-electric car for its employees to travel to client sites. The base price of the car is ₹32,78,000, but the company is eligible for a government subsidy of ₹7,50,000, which reduces the effective purchase cost. The firm is evaluating two options: it can lease the car by paying ₹4,20,000 per year at the end of each year, along with an initial cost of ₹2,50,000, or it can purchase the car and a home charging station. The home charging system costs ₹2,20,000. The company expects that after 3 years, the car and charging station can be sold for 40% of the car's base price. Using an interest rate of 10% per year, should the company purchase or lease the vehicle? Use annual worth approach. (3)

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2C) E&T Construction Company is bidding to build a bridge across a river on a major highway connecting two metropolitan cities in the coastal area. Therefore, severe cyclones are expected to hit the project area every six years, causing structural damage, adding on to the costs. According to the tender, the company must construct and maintain the bridge over the years forever. Determine the total amount for which the company must bid now (assume interest rate of 8%). The project engineers have estimated the following costs: (3)

Cost Item	Amount (₹)
Design & Construction	5,00,00,000
Resident Relocation	30,00,000
Annual Road repairs	10,00,000 for first 10 years, then onwards it increases to 15,00,000
Cyclone Structural Repairs	80,00,000 at the end of every 6 years

8%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099
9	1.999	.5002	.0801	.1601	12.488	6.247	3.491
10	2.159	.4632	.0690	.1490	14.487	6.710	3.871
11	2.332	.4289	.0601	.1401	16.645	7.139	4.240
12	2.518	.3971	.0527	.1327	18.977	7.536	4.596
13	2.720	.3677	.0465	.1265	21.495	7.904	4.940
14	2.937	.3405	.0413	.1213	24.215	8.244	5.273
15	3.172	.3152	.0368	.1168	27.152	8.559	5.594

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3A) A company is starting pilot plant operations for a new process it has developed. One optional piece of equipment that can be purchased is a heat exchanger unit. The company finds that a similar unit is available now for ₹26.50 lakhs. This unit can also be used later in other company projects. It is estimated that the heat exchanger will have a resale value of ₹30.74 lakhs at the end of 8 years. The high resale value is mainly because the current purchase price is considered a rare bargain. If the company expects a 10% annual rate of return, what annual benefit (in ₹) must the equipment generate to make the purchase worthwhile? (2)

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476
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5	1.611	.6209	.1638	.2638	6.105	3.791	1.810
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) A manufacturing firm is evaluating two production lines - the existing one (A) and a proposed new one (B). (4)

Particulars	Line A	Line B
Cost of equipment	₹3,00,000	₹4,20,000
Annual sales	₹5,00,000	₹5,00,000
Annual operating cost	₹3,00,000	₹2,60,000
Life	5 years	5 years
Salvage value	Nil	Nil

The MARR=10%. Using the incremental rate of return approach, calculate the incremental IRR and determine whether to replace Line A with Line B.

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388

20%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.200	.8333	1.0000	1.2000	1.000	0.833	0
2	1.440	.6944	.4545	.6545	2.200	1.528	0.455
3	1.728	.5787	.2747	.4747	3.640	2.106	0.879
4	2.074	.4823	.1863	.3863	5.368	2.589	1.274
5	2.488	.4019	.1344	.3344	7.442	2.991	1.641
6	2.986	.3349	.1007	.3007	9.930	3.326	1.979
7	3.583	.2791	.0774	.2774	12.916	3.605	2.290
8	4.300	.2326	.0606	.2606	16.499	3.837	2.576
9	5.160	.1938	.0481	.2481	20.799	4.031	2.836
10	6.192	.1615	.0385	.2385	25.959	4.192	3.074
11	7.430	.1346	.0311	.2311	32.150	4.327	3.289
12	8.916	.1122	.0253	.2253	39.581	4.439	3.484

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3C) A contractor wants to determine whether to purchase a used Caterpillar mini excavator or a Toro powered rotary tiller for servicing irrigation ditches in California. The details for the two options are as follows: (4)

Option 1: Caterpillar Mini Excavator	Option 2: Toro Powered Rotary Tiller
- Initial cost: ₹26,500 - Salvage value after 10 years: ₹9,000 - Fixed annual costs (insurance, license, etc.): ₹18,000 per year - Operator cost: \$15 per hour - Maintenance: \$1 per hour - In 1 hour, 0.15 mile of ditch can be prepared.	- Initial cost: ₹1,200 - Useful life: 5 years, no salvage value - Operator cost: 2 workers × \$11 per hour each - Operating cost: \$1.20 per hour - The two workers can prepare 0.04 mile of ditch in 1 hour.

The contractor's minimum acceptable rate of return (MARR) is 10% per year. Determine the number of miles of ditch per year, the contractor would need to service for the two options to break even.

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381
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7	1.949	.5132	.1054	.2054	9.487	4.868	2.622
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4A) The operations manager at GreenFleet Logistics Pvt. Ltd. is reviewing whether to continue using an existing delivery vehicle or replace it with a newer model. The current vehicle, purchased six years ago, has a market value of ₹3.5 lakhs now. If retained for another five years, its annual operation and maintenance (O&M) cost will be ₹50,000 in the first year, increasing by ₹20,000 each year thereafter. The vehicle will have a salvage value of ₹1.25 lakhs at the end of five more years. (4)

Two replacement options are available:

Particulars	Option A – EcoDrive Model X	Option B – VoltRide Hybrid Z
Purchase Price	₹8,00,000	₹9,50,000
Annual O&M Cost	₹30,000 (constant)	₹25,000 (constant)
Salvage Value after 5 years	₹5,00,000	₹6,00,000
Trade-in (Exchange) Value for Old Vehicle	₹3,70,000	₹4,00,000

Using the cashflow approach in replacement analysis, determine whether it is financially more viable to retain the existing vehicle for another five years or replace it with either EcoDrive Model X or VoltRide Hybrid Z.

Assume the MARR of the company to be 10%.

10% n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) The finance manager of Nova Textiles Ltd. is analyzing the company's short-term financial position (4) and efficiency for the year ending March 31, 2025. The following information is available:

Financial Information:

- Current Ratio = 2.5
- Quick Ratio = 1.5
- Inventory Turnover Ratio = 6 times (assuming closing inventory is equal to average inventory)
- Gross Profit Ratio = 25% of Sales
- Net Sales = ₹18,00,000

Current Assets	Current Liabilities
• Cash in Hand = ₹50,000 • Bills Receivable = ₹1,00,000 • Debtors = ₹2,50,000	• Outstanding Expenses = ₹30,000 • Short-term Notes Payable = ₹70,000

Using the given information and applying ratio relationships, determine:

- The value of Inventory
 - The value of Sundry Creditors
 - The total value of Current Assets and Current Liabilities
- The Cost of Goods Sold (COGS)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4C) A company is evaluating two investment projects, Project A and Project B, under three possible (2) future economic conditions.

The possible cash inflows (₹ in lakhs) and their probability are as follows:

Economic Condition	Probability	Project A	Project B
Good	0.3	15	20
Normal	0.5	12	10
Poor	0.2	6	2

- Calculate the Expected Monetary Value (EMV) for each project. Identify which project should be selected based on the expected value criterion.
- Explain how the concept of Expected Value helps in managing project risk when future economic conditions are uncertain.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) Mithra Foods & Packaging, a small FMCG vendor in Coimbatore, is planning to buy a semi-automatic pouch (4)

filling machine to increase its packaging capacity. The machine costs ₹9,00,000 and is expected to last for 6 years. The annual production is estimated to be 30,000 pouches per year beginning from the first year onwards. The annual operating and maintenance cost will be ₹90,000 for the first two years, and from the third year onward, this cost will increase by ₹15,000 every year. In addition, an overhaul cost of ₹75,000 will occur once every three years, including the last year of operation. The machine is expected to have a salvage value of ₹90,000 at the end of its life. If the company's minimum attractive rate of return (MARR) is 10% per year, determine the minimum selling price per pouch that Mithra Foods & Packaging should set to cover all costs and have zero annual worth over the 6-year period.

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	<i>P/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 5B) Haixun Industrial Technologies (USA) Ltd., operating in Texas, is evaluating the purchase of a new automated metal processing unit costing \$240,000. The equipment will incur annual operating costs of \$36,000 in the first year, increasing by a gradient of \$11,000 per year due to rising maintenance and labor expenses. Its salvage value declines by 20% each year from the initial purchase price for the first three years and from the fourth year onwards the salvage value remains constant at \$50,000. The company's Minimum Attractive Rate of Return (MARR) is 12% per year. Determine the economic service life of the machine and its subsequent minimum annual cost value. (4)

12%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	<i>P/G</i>
1	1.120	.8929	1.0000	1.1200	1.000	0.893	0	0
2	1.254	.7972	.4717	.5917	2.120	1.690	0.472	0.797
3	1.405	.7118	.2963	.4163	3.374	2.402	0.925	2.221
4	1.574	.6355	.2092	.3292	4.779	3.037	1.359	4.127
5	1.762	.5674	.1574	.2774	6.353	3.605	1.775	6.397
6	1.974	.5066	.1232	.2432	8.115	4.111	2.172	8.930
7	2.211	.4523	.0991	.2191	10.089	4.564	2.551	11.644
8	2.476	.4039	.0813	.2013	12.300	4.968	2.913	14.471
9	2.773	.3606	.0677	.1877	14.776	5.328	3.257	17.356
10	3.106	.3220	.0570	.1770	17.549	5.650	3.585	20.254

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) Delta Traders had credit purchases of ₹12,00,000 during the year, with opening creditors of ₹2,00,000 and closing creditors of ₹1,50,000. Calculate the Creditors Turnover Ratio and the Average Payment Period (assuming 360 days in a year). (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	HUM_3021:ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	10

3	3	CO 3	2
4	4	CO 4	24
			Total 50

* NA - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	HUM_3021:ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	23(46%)
4	Analysis	8	27(54%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)